

NYC Airbnb Market Segmentation Analysis



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Why Study NYC Airbnb?



- Short-term rentals (e.g., Airbnb) have reshaped hospitality.
- NYC: A global tourism hub with a complex Airbnb market.
- Business Question:

How is the NYC Airbnb market segmented, and what drives these segments?

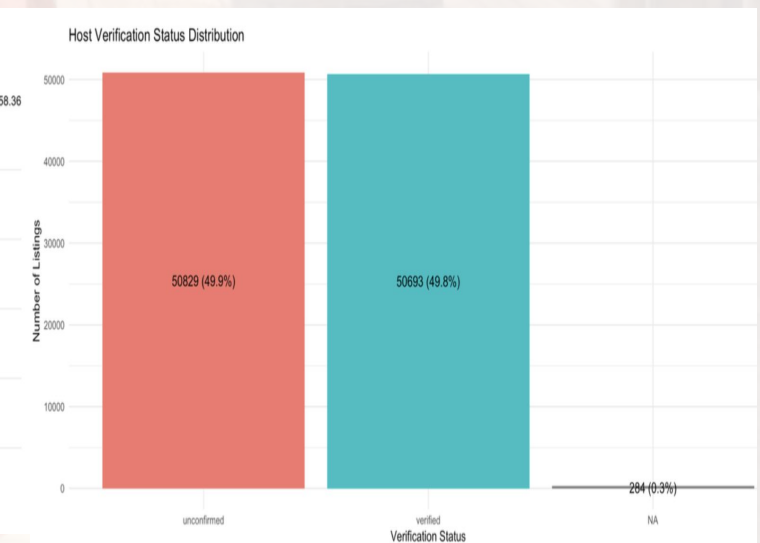
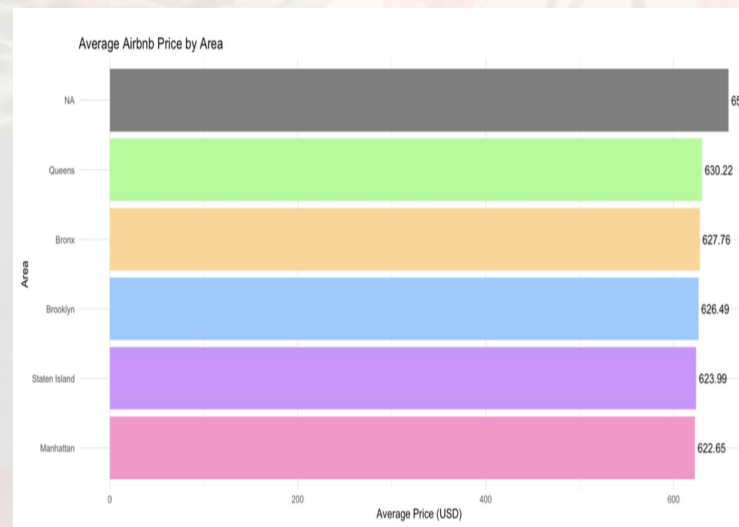
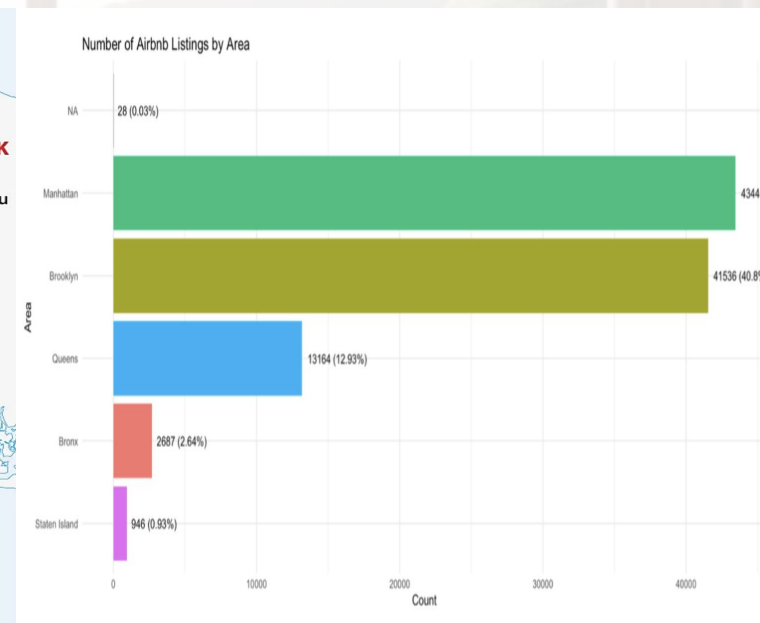


What We Expected

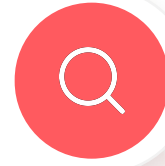
- Hypothesis: Listings cluster by location (borough) and price
- Based on real estate mantra: “Location, location, location.”

What EDA Showed

- Geographic Concentration: 83.5% of listings in Manhattan (42.7%) and Brooklyn (40.8%)
- Prices consistent across boroughs (~\$622-\$658)
- Verification Balance: Equal split of verified/unverified hosts



Methodology



Data

2022 Kaggle Airbnb Open Data
(price, minimum nights, reviews, etc.)

Tools

R (tidyverse, e1071, cluster, factoextra)

Techniques

- Exploratory Data Analysis (EDA) for trends
- K-means Clustering to identify segments ($k=3$)
- Support Vector Machine (SVM) to validate clusters (85% accuracy)



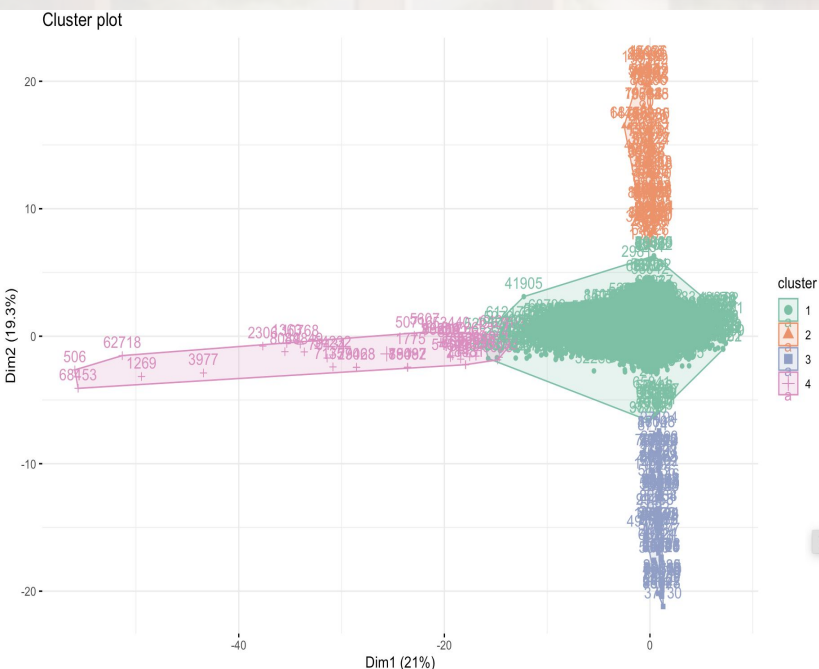
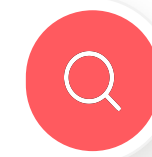
Key Findings-Cluster Characteristics



Segment	Price	Min. Nights	Reviews/Month	Total Reviews
Long-Term Stays	\$618	12.1	0.896	18.4
High-Turnover Stays	\$635	3.07	4.04	116
Moderate-Turnover Stays	\$624	5.71	0.827	14.5

Insight: Minimum nights and review patterns, not price or location, drive segmentation

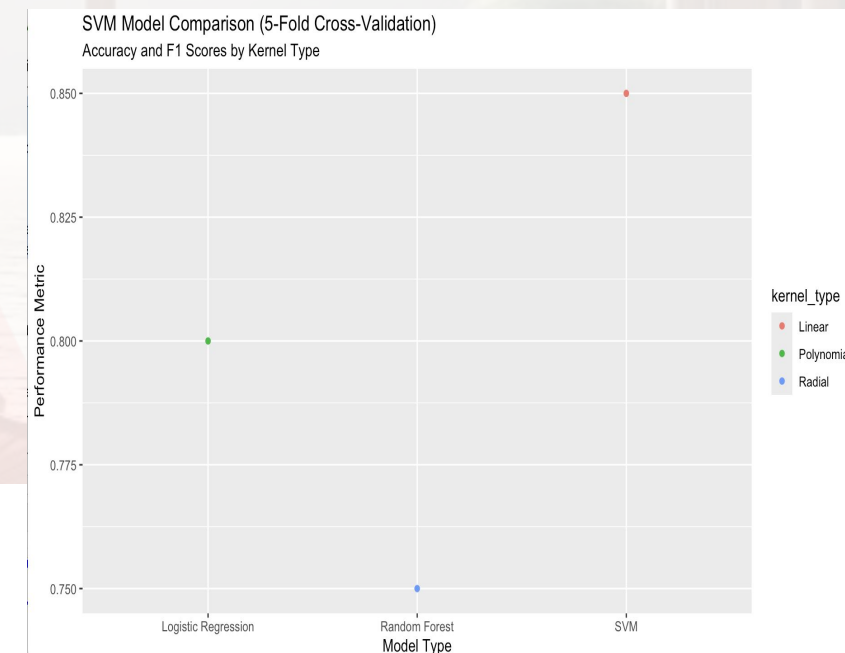
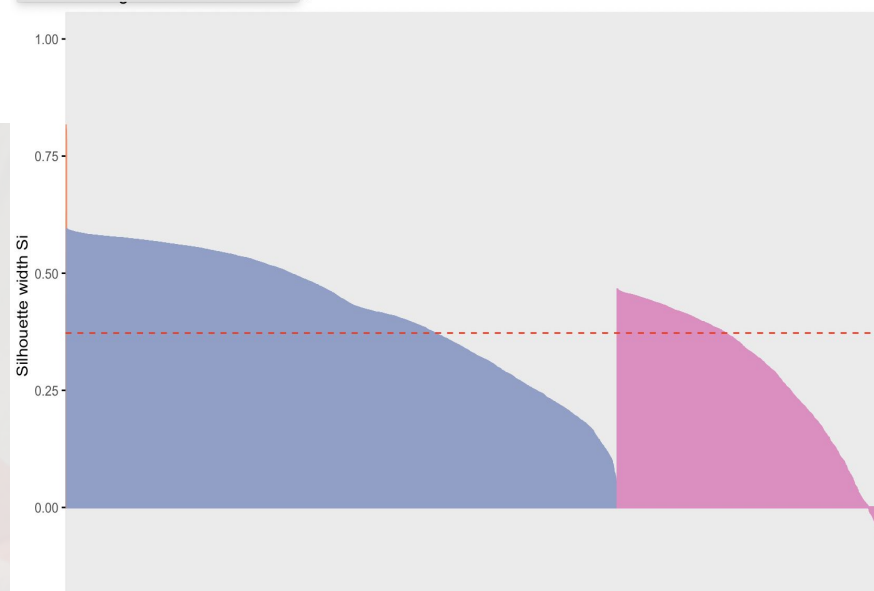
Visualizations



Silhouette Plot

**Moderate clustering structure
(width = 0.37).**

Clusters silhouette plot
View a larger version of the plot in a new window 37



Cluster Scatter Plot

- Shows 3 main clusters (PCA: Dim1 21%, Dim2 19.3%).
- Long-term (blue), high-turnover (orange), moderate (green).

SVM Decision Boundaries

**Review frequency and
minimum nights as key
differentiators.**

Implications & Recommendations



Hosts



- **Long-Term: Flexible policies for extended stays.**
- **High-Turnover: Fast responses, tourist-friendly amenities.**

Platform



Add segment-based search filters

Investors



Target properties for high-turnover or long-term segments.

Future Research & Conclusion



What's Next?

- Analyze host professionalism (e.g., response time)
- Study geographic density and seasonal trends
- Explore the small “super-host” cluster (44 listings)

Takeaway

- NYC Airbnb market is defined by **operational strategies**, not location or price
- Strategic alignment = success for hosts, platforms, and investors





Thank You